Appendix 2 exercises

1. For A of size 2×3, B of size 3×2, and C of size 2×2, determine whether $A + B$, AB, BA, $A^T C$ exist and give resulting dimensions.
2. Let $A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}$, $B = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$, $C = \begin{pmatrix} 1 & -1 \\ 2 & 0 \end{pmatrix}$. Compute AB, BA, $A(B + C)$, $AB + AC$ and check distribution.
3. Use cofactor expansion along any row to compute $\det \begin{pmatrix} 2 & 1 & 0 \\ 0 & 3 & 1 \\ 1 & -1 & 2 \end{pmatrix}$.
4. Determine whether $A = \begin{pmatrix} 4 & 1 \\ 7 & 2 \end{pmatrix}$ is invertible. If so, find its inverse and verify $A^{-1}A = I_2$.
5. Prove $(AB)^T = B^T A^T$, and give invertible 2×2 matrices with $AB \neq BA$.
6. Solve $2x + y = 5, x + 3y = 7$ by inverse or Cramer. If the first equation becomes $4x + 2y = 10$ while the second stays unchanged, is the solution still unique? Can the original inverse be reused without adjustment?
7. For $A = \begin{pmatrix} 1 & 2 \\ 2 & 4 \end{pmatrix}$, eliminate the augmented systems with $b = (3, 6)^T$ and $(3, 7)^T$. Give all solutions or the contradiction. Why does $\det A = 0$ not decide consistency?
8. Prove the adjugate identities and equivalence of invertibility with nonzero determinant. Give a size-two counterexample to determinant additivity, and explain $\det(cA) = c^n \det A$.
9. Let $A = (1, 0)^T$ be 2×1, and $B = (1, 0)$ be 1×2. Compute BA, AB, identify left and right inverses, and explain why $BA = I_1$ does not imply $AB = I_2$.
10. Prove an integer matrix has an integer-entry two-sided inverse iff its determinant is ± 1. Can determinant two still allow real invertibility? Give an example.
11. Let $N = \begin{pmatrix} 0 & 1 \\ 0 & 0 \end{pmatrix}$ and $M = N^T$. Compute N^2, M^2, NM, MN, and compare $(N + M)^2$ with $N^2 + 2NM + M^2$. Under what condition may the latter be used?

References and attribution

Lin Cing, handwritten linear algebra notes. Ralph P. Grimaldi, Discrete and Combinatorial Mathematics, 5th ed., Appendix 2.

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https://github.com/xinhi6716/computer-science-self-study